NETWORK GROWTH MODELS

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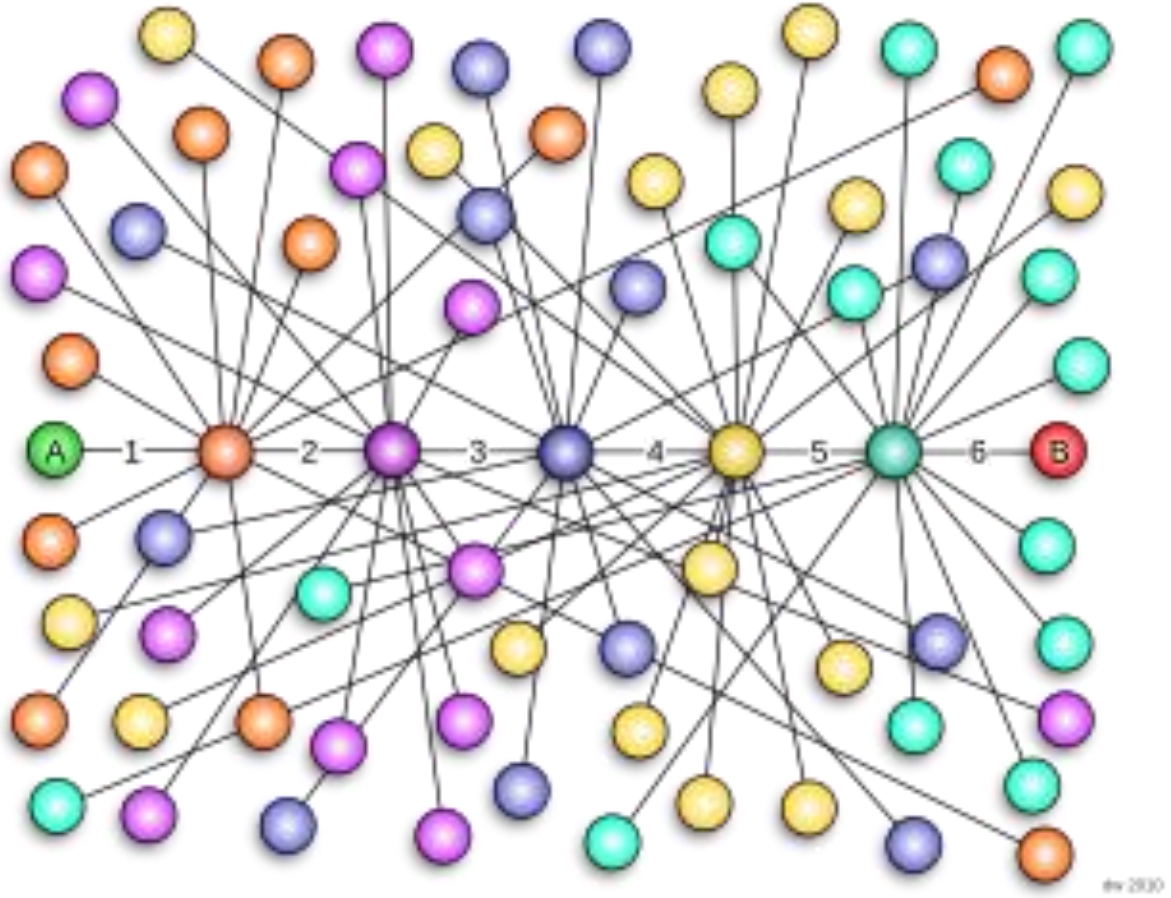
Need for network growth models

- Real networks are very huge.
- To develop new models/ algorithms we need a smaller version of that network
- So we create synthetic network
- But that synthetic network must exhibit all the properties of the real-world network
- Then only our developed algorithms will work when deployed in the real network.

Properties of real world networks

- Small-world property
- High average local clustering coefficient
- Scale free property

Small world property



Small world property

Milgram's Small world experiment

Milgram typically chose individuals in the U.S. cities of Omaha, Nebraska, and Wichita, Kansas, to be the starting points and Boston, Massachusetts, to be the end point of a chain of correspondence. These cities were selected because they were thought to represent a great distance in the United States, both socially and geographically

Source: Wikipedia

Small-world property

- Stanley Milgram's 1967 "Small World" experiment was a landmark study in social psychology that sought to measure the "distance" between any two people in society. It popularized the concept we now know as **six degrees of separation**.
- **The Experiment Setup**
  - Milgram wanted to see how many intermediate acquaintances it would take to connect two random strangers.
  - **The Starting Point:** Approximately 296 participants were recruited from Omaha, Nebraska, and Wichita, Kansas.
  - **The Target:** A stockbroker living in Sharon, Massachusetts, who worked in Boston.
  - **The Task:** Each participant was given a folder and told to get it to the target. However, they could only send it to someone they knew **personally** on a first-name basis who might be "closer" to the target.

Small-world property

- Out of 296, only 64 letters reached their destination.
- When they plotted intermediate path length it was 5.4
- Minimum path length 1 and maximum 11

Is today's online world even smaller than Milgram's small-world?

- Is today's online world even smaller than Milgram's small-world?
- In the year 2008, a study by Microsoft showed that the average chain of contacts between users of Microsoft Messenger service was 6.6 people (Leskovec and Horvitz 2007).
- In May 2011, Facebook, with its 721 million active users and 68 million friendship links, showed that the average distance between two random Face-book users is 4.74 (with 3.74 intermediaries). This study coined "four degree of separation" (Backstrom et al. 2012).
- The same experiment was repeated in 2016 with 1.59 billion people active on Facebook, and the average distance they observed was 4.57, corresponding to 3.57 intermediaries.
- Within the U.S. people were connected to each other by an average of 3.46 degrees. Facebook claimed that the "degrees of separation" have shrunk over time due to exponential growth in the number of nodes and interconnections.

Small-world property

- Small-world property
- A network G is said to follow the small world property is the average path length of network is logarithmically proportional to the network size:

$$l_G \propto \log(|G|)$$

High average local clustering coefficient

High value of local clustering coefficient indicates that at an individual level, the neighbours of a node tend to be highly connected with each other. For example, in a protein–protein interaction network, clustering coefficient of 0.73 was obtained from a network of size 100 (Miller et al. 2007). Ugander et al. (2011) showed that the Facebook social network has an average local clustering coefficient of 0.14, which is unexpectedly high for a graph of 150 million nodes.

Scale free property

- Majority of the real-world networks exhibit scale invariance on the degree distribution as the degrees of nodes follow power law distribution as defined below
- Power Law:
  - The power law is a relationship between two entities such that one entity varies in a power of the other. Mathematically, if two entities, x and y, are related by the formula

$$y \propto x^{-b} \Longrightarrow y = \alpha x^{-b}$$

  - Where b is a real number and α is a constant

Scale free property

- **A function which follows power law is scale-invariant.**
- This is so because when we scale the input by a constant c, the function remains proportional to the original function.
- In other words, power laws have the same functional form at all scales.
- The power law remains unchanged (other than a multiplicative factor) when rescaling the independent variable x.
- For example, let $y = f(x) = \alpha x^{-b}$. If we scale x by c, then

$$f(cx) = \alpha(cx)^{-b} = c^{-b} f(x) \implies f(cx) \propto f(x)$$

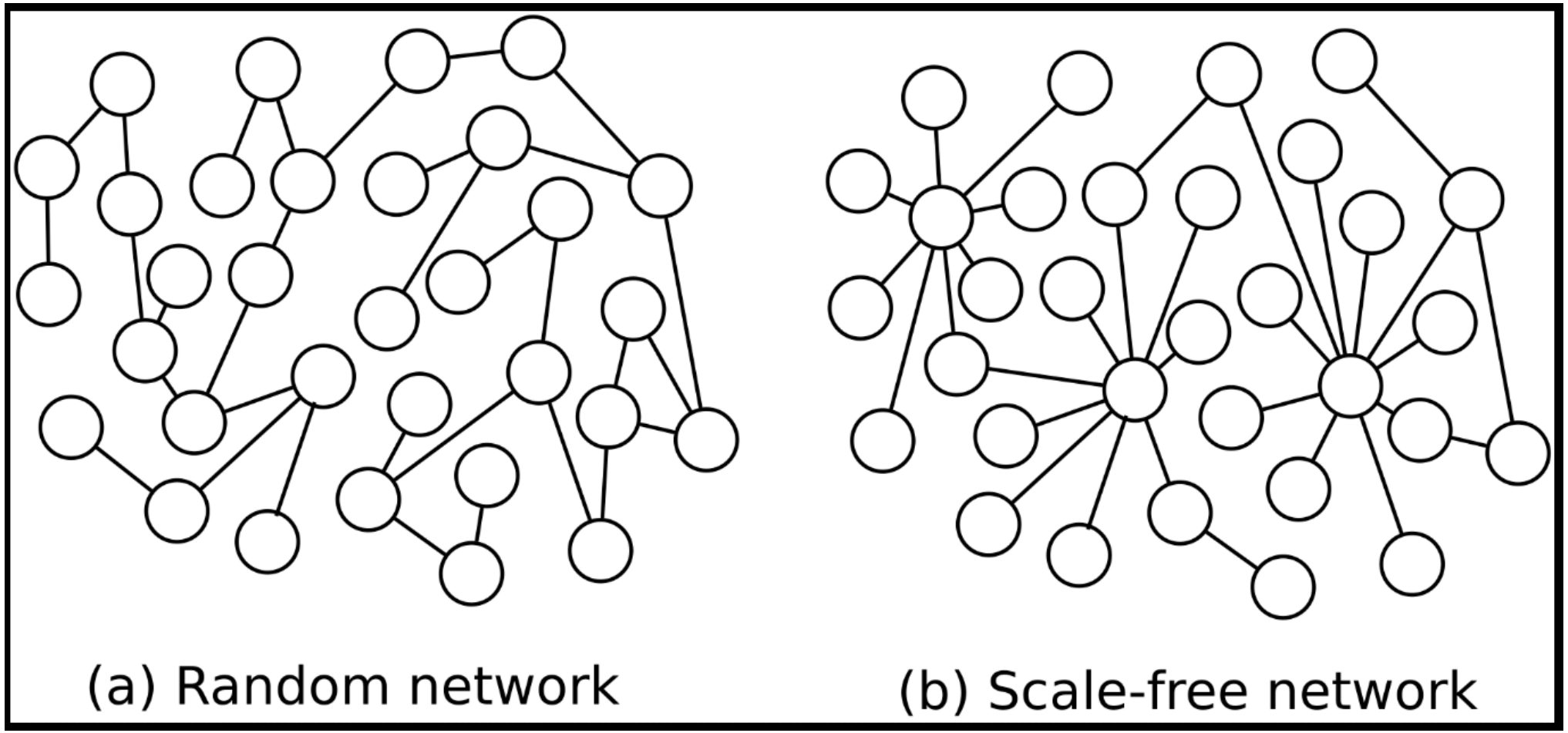
Scale free property

- **Power law or Pareto distribution**
  - Power law, in statistics, plays a vital role to quantify the relation between two variables a relative change in the value of one variable leads to the proportional change in the value of other variable, independent of the initial values of both the variable, we say that one varies as a power of another.
  - A simple example is the area of a square in terms of the length of its side if the length is doubled, the area is multiplied by a factor of four.
  - Another example of power law is the famous Pareto Principle (or 80/20 rule Marshall (2013), the law of the vital few, or the principle of factor sparsity) in economics, which states that 80% outcomes are results of 20% of the causes.
  - For example, in a company, 80% profits are caused by 20% employ-ees, in a country, 80% properties are consumed by 20% citizens; in a business, 80% of sales come from 20% of clients (Newman 2005).
  - Therefore, the power law principle is also called Pareto distribution.
  - Pareto distribution was named after the famous Italian economist and philosopher, Vilfredo Federico Damaso Pareto, who first observed the 80/20 rule or the top 1% rule in the wealth distribution among citizens in Italy 80% of the money was eared by 20% of the population.

Scale free property

- In the world of network science, a **scale-free network** is one where a few individuals have a massive number of connections, while the vast majority have very few
- If you look at a typical "random" network, most people have roughly the same number of friends (following a bell curve). In a scale-free network, however, the distribution follows a **Power Law**.

Scale free property



Scale free property

1. A **scale-free network** is a network whose degree distribution follows a power law, at least asymptotically. That is, the fraction *P*(*k*) of nodes in the network having *k* connections to other nodes goes for large values of *k* as

$$P(k) \sim k^{-\gamma}$$

Where γ is a parameter whose value is typically in the range $2 < \gamma < 3$

This math essentially tells us that "hubs" (super-connected individuals) aren't just rare—they are a fundamental, predictable part of the system's structure.

Scale free property

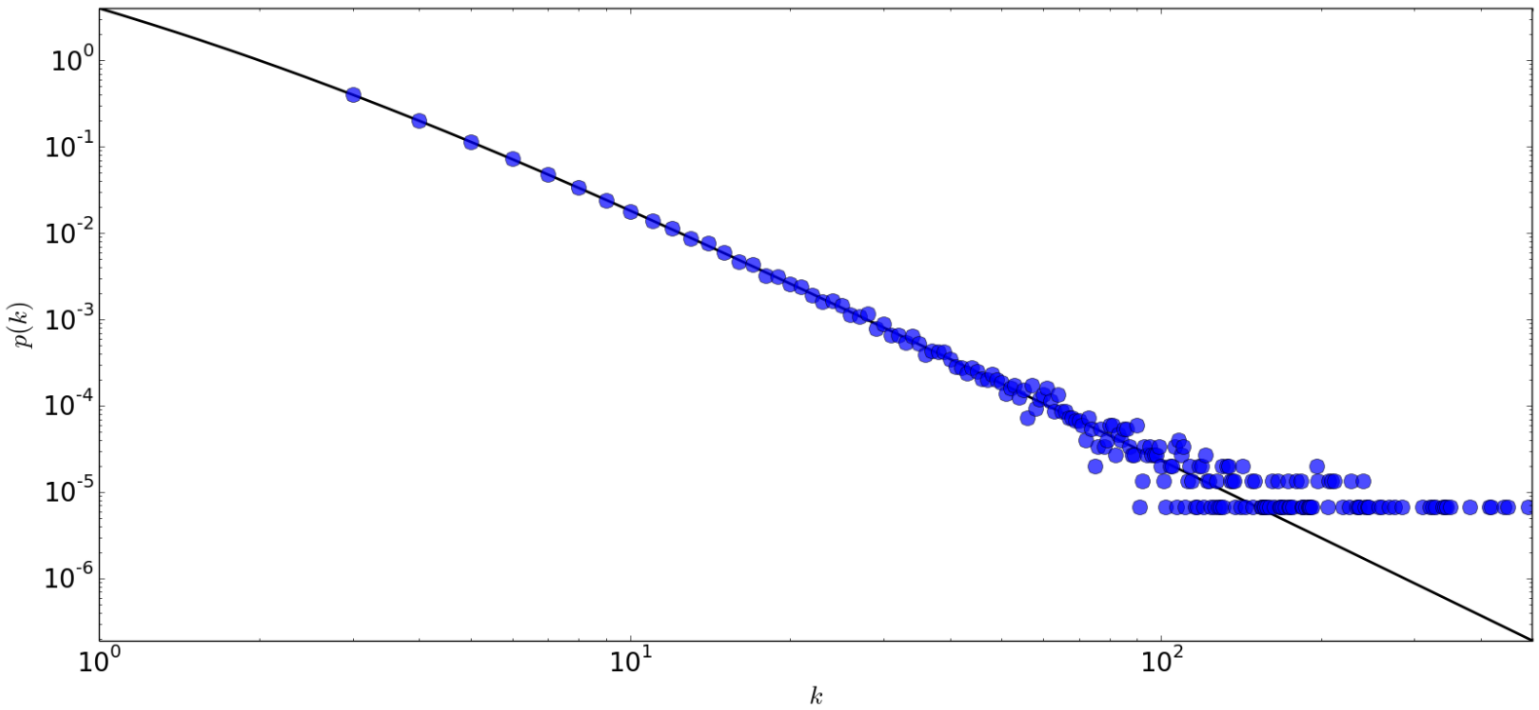
2. The Presence of "Hubs"

- Think of celebrities on X (formerly Twitter) or LinkedIn "Lions." While you or I might have a few hundred followers, a hub has millions. These hubs act as the "glue" of the social network, significantly shortening the path between any two random people.

3. Growth and Preferential Attachment

- Scale-free networks don't just appear; they grow. They usually follow a principle called **Preferential Attachment**, or more colloquially, "the rich get richer."
- When a new person joins a social network, they are more likely to connect to someone who is already well-known (a hub) than to someone with no followers.

Scale free property



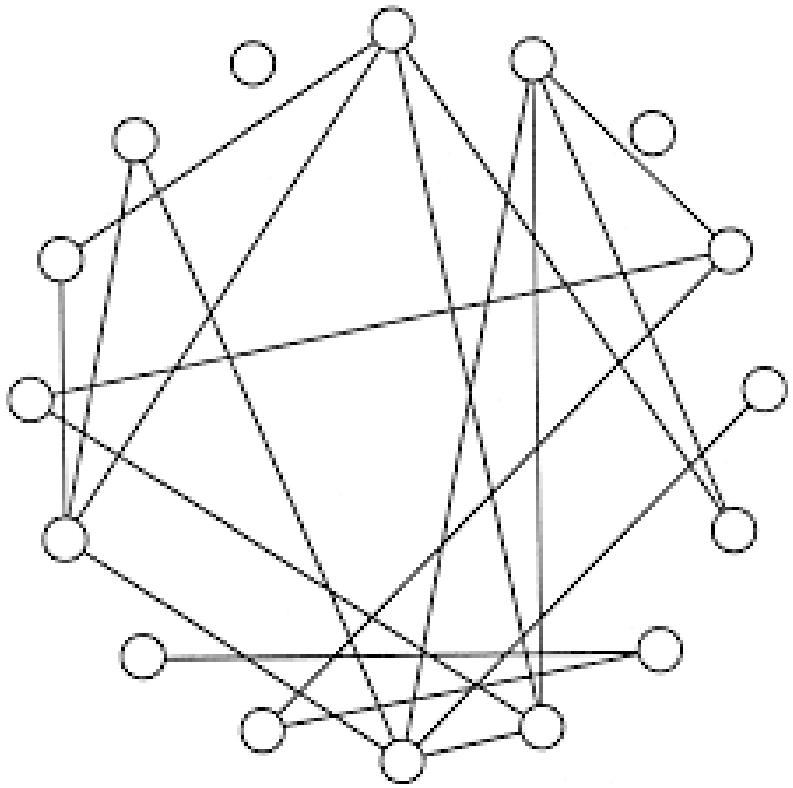
Degree distribution for a network with 150000 vertices and mean degree = 6 created using the Barabási–Albert model (blue dots). The distribution follows an analytical form given by the ratio of two gamma functions (black line) which approximates as a power-law.

Random Network Model (Erdos-Renyi Model)

Random Network Model

- Random network model is one of the simplest proposed network models, introduced by **Erdős and Renyi** (1960).
- A random network G is a network of N nodes such that any random pair of nodes is linked by an edge with a fixed probability.
- There are various random network models; different models yield different probability distributions on the resulting random networks.
- Two popular variants of the model are:
  - G(N, K) Model: In this variant of the model, originally proposed by Paul Erdős and Alfréd Rényi, a network is chosen uniformly at random from the set of all possible networks with N nodes and K edges (Erdős and Rényi 1959, 1960, 1964). Figure 3.3 shows a toy example of a random network.
  - G(N, p) Model: In this variant of the model, originally proposed by Edgar Gilbert independently with Erdős and Rényi, there are N nodes in the network and any random pair of nodes has a probability p of being adjacent independently with any other pair of nodes in the network (Gilbert 1959).

Random Network Model



An instance of $G(16,1/7)$ network

Random Network Model

- It can be shown analytically that both the above variants behave identically in the limiting case, and hence both are called the random network model.
- The G(N, p) variant is considered standard now, and is widely used representation since number of links varies random networks.
- Using this representation, we could define the probability P(K) of having K edges in a random network.

Random Network Model

- Defining the probability P(K) of having K edges in a random network:
  - In a network G, there are N nodes
  - Maximum possible number of edges $=N_{C_2}$
  - We need to choose K edges
  - This could be done in $(_{N_{C_2}})_{C_K}$ ways
  - According to the model assumption, the probability that there is an edge between a random pair of node =P
  - Therefore, probability of existence of K random edges = P^K
  - And, probability of absence of $(N_{C_2}$- K) edges = $(1-P)^{(N_{C_2}-}$ K)

Random Network Model

- As above two events have taken place simultaneously, we have
- P(K) = $(_{N_{C_2}})_{C_K} . P^K . (1-P)^{(N_{C_2}-}$ K)
- Note that above expression is an instance of binomial distribution
- P(K) = $N_{C_x} . P^x . (1-P)^{N-x}$
- Mean of binomial distribution = N.P
- Using this formula:

- **Average number of edges <K>** in a random network G(N,P) = $N_{C_2} . P$
  - <K> = $\frac{N(N-1)}{2} . P$

Random Network Model

- Similarly, the **average degree <d>** in a random network is computed as follows:
- $<d> = \frac{Sum\ of\ degrees\ of\ all\ nodes}{Number\ of\ nodes} = \frac{2 \times Total\ edges}{Number\ of\ nodes} = \frac{2<K>}{N}$
- $< d > = \frac{P.N.(N-1)}{N} = P.(N-1)$
- Note that both <K> and <d> vary linearly with P.
- As we increase P from 0 to 1, K increases from 0 to $K_{max} = \frac{N.(N-1)}{2}$, and <d> increases linearly from 0 to N-1.
- Hence, larger the value of P, denser the resulting random network.

Random Network Model

- **Degree distribution of a random network:**
  - To determine the degree distribution of a random network, we closely examine one of the nodes of the random network. The aim is to find the probability P(K) that a particular node has a degree K.
  - For the node to have degree K, the node must be adjacent to K other nodes directly
  - Clearly, we can select K nodes at random from remaining N - 1 nodes in $N - 1_{C_K}$ possible ways.
  - The probability of the existence of edges between the selected K node pair is P^K .
  - On the contrary, the probability of the absence of remaining (N-1)-K edges is $(1 - P)^{(N-1)-K}$
  - Since the aforementioned two events are independent, we have the following result:
- P(K) = $N - 1_{C_K}.P^K.(1 - P)^{(N-1)-K}$
- Figure in next slide shows the degree distribution of random networks with varying number of nodes.

Random Network Model

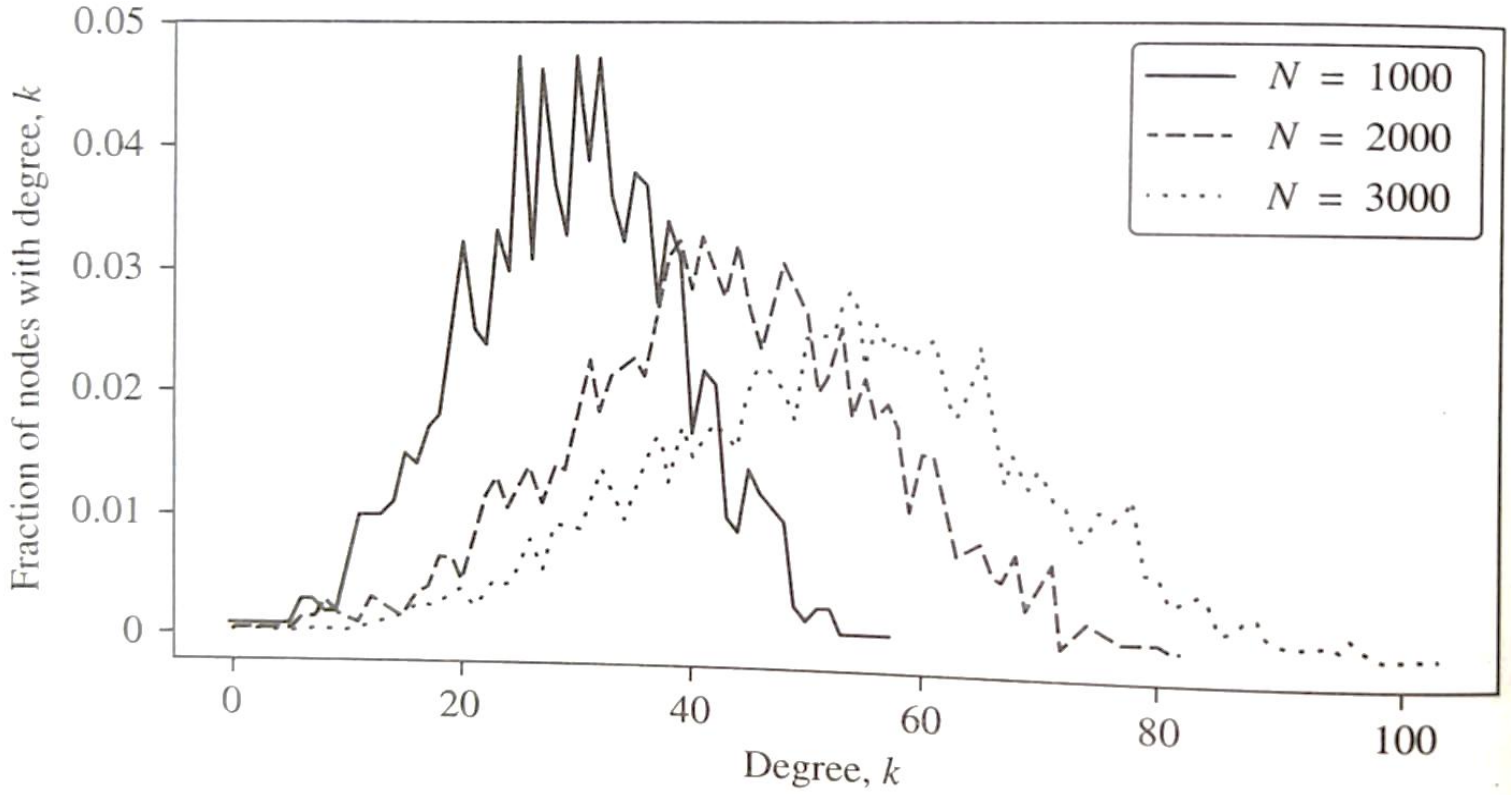
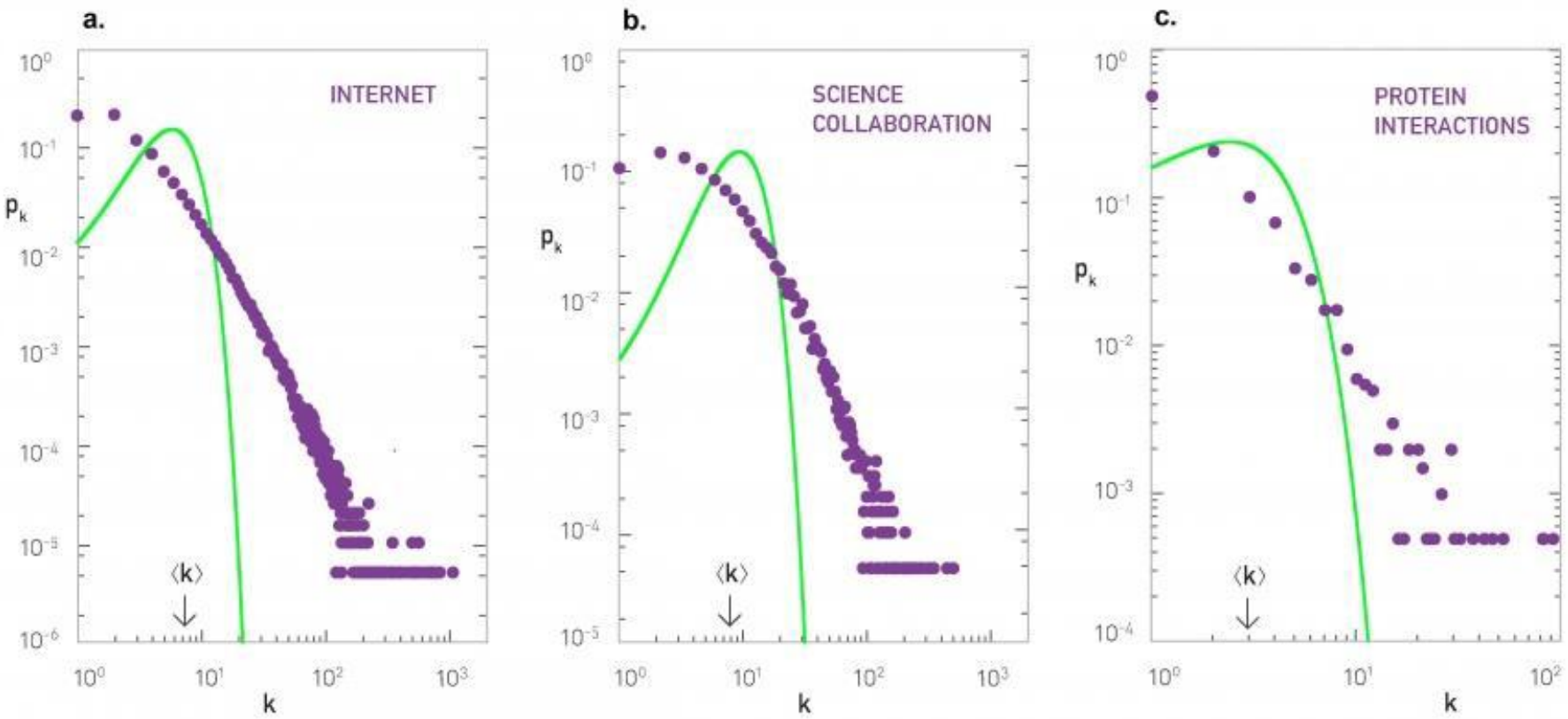


Figure: Degree distribution of random networks with varying number of nodes

Random Network Model



Real-life networks are often scale-free; however, Erdős-Rényi Networks are not
http://networksciencebook.com/chapter/3#not-poisson

Random Network Model

- **Binomial to Poisson distribution:**
  - As N → ∞, we can approximate a binomial (N,P) into Poisson distribution. Applying the Poisson approximation to the binomial distribution, we get
  - $P(K) = e^{-<d>} \times \frac{<d>^K}{K!}$
- **Average Path Length:**
- Let us examine a node in the random network individually. Refer Figure (next slide), on an average, a node in the random network has <d> neighbours (path distance = 1), where <d> is the average degree in the random network.
- Similarly, there are <d> ^{2}nodes at a distance of 2 from the node.
- Extrapolate this till the last level, that is, l_{max}(maximum path length in random network), we see $< d >^{l_{max}}$ nodes.
- Adding the average degree of nodes at each level, we get
- $1 + <d> + <d>^2 + <d>^3 + \ldots + < d >^{l_{max}} = N$

Random Network Model

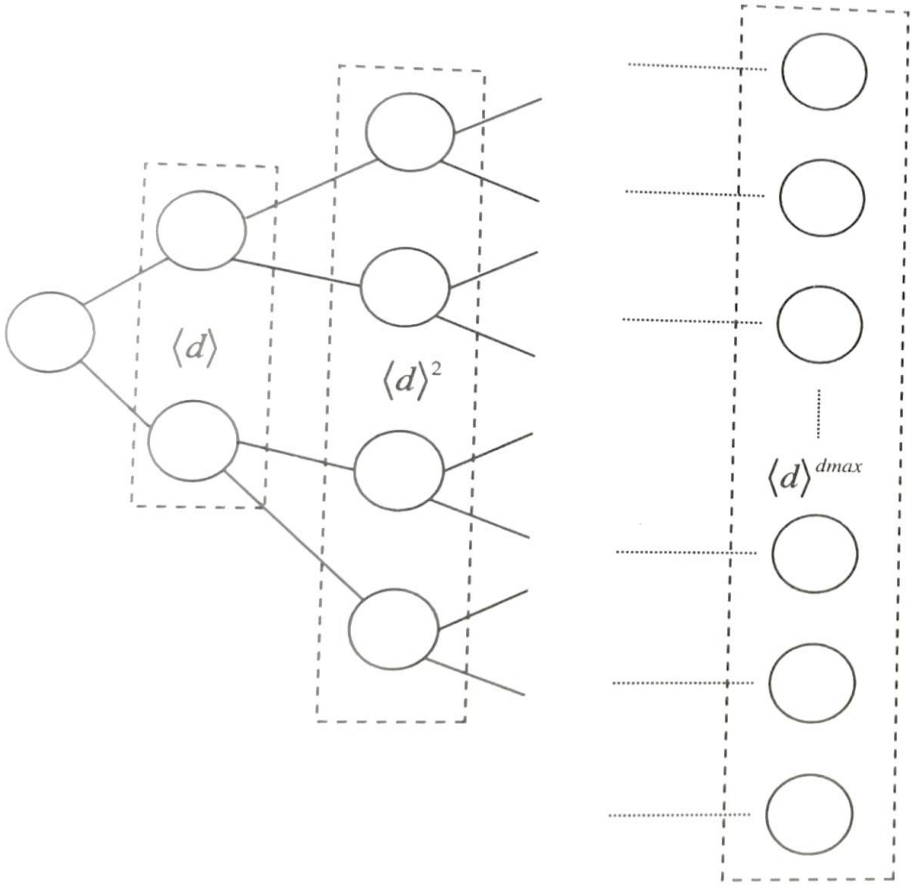


Figure: A depiction of a random network in tree format. As the level increases, the number of nodes increases by a multiple of <d>

Random Network Model

- Applying sum on this geometric series, we get

$$\frac{<d>^{l_{max}+1}-1}{<d>\ -1} = N$$

When <d> >> 1, it is approximated as

$$< d >^{l_{max}} = N \Rightarrow\ l_{max} = \frac{\log N}{log<d>}$$

Approximating further, we obtain

$$< l > = \frac{logN}{log<d>}$$

$$< \boldsymbol{l} > \propto \boldsymbol{logN}$$

Average path length in a random network is proportional to the logarithmic of the system size. Hence, the random networks follow **small world property**.

Random Network Model

- **Clustering Coefficient:**
  - According to the definition of local clustering coefficient, the number of possible edges between the neighbours of a node in a random network is $< d >_{C_2}$.
  - Now the actual number of edges between the neighbours of node v_i in a random network G would simply be the expected number of edges between <d> nodes from random network, which is P. $< d >_{C_2}$
  - Applying the formula of local clustering coefficient, we get

$$C_i = \frac{P.<d>_{C_2}}{<d>_{C_2}} = P = \frac{<d>}{N}$$

Random Network Model

- Therefore, the local clustering coefficient for a random network is inversely proportional to the network size (for a fixed P). Thus, for a fixed P, a larger random network would result in a smaller clustering coefficient or into a sparse random network.
- It is also interesting to note that in a random network, the local clustering coefficient of a node does not depend on the degree of the node.

Random Network vs. Real-World Network

- **Small-world property:** As we saw earlier, random networks do follow the small-world property. Real-world networks typically follow small-world property, too.
- **Power law:** The degree distribution of random networks does not follow power law. This is unlike real-world networks that are scale-free in nature.
- **Clustering coefficient:** In a random network, as we increase the network size, the local clustering coefficient decreases. However, in real-world networks like the online social networks, we see that nodes with high local clustering coefficient exist in enormously large graphs as well.
  - For example, there are echo chambers in large social networks such as Facebook, wherein people with a similar view on a topic are all friends with each other.

Ring Lattice Network Model

Ring Lattice Network Model

3.3 RING LATTICE NETWORK MODEL

Ring Lattice Network

A ring lattice network consisting of N nodes labelled $0, 1, 2, \ldots, N-1$ arranged in circular order, is a network such that every node is connected to k nodes, with $k/2$ nodes on each side of its position in the ring. The position is maintained such that there is an edge $\langle i, j \rangle$ if and only if $0 < |i-j| \text{mod}(N-1-k/2) \leq k/2$. In short, every node is connected to immediate $k/2$ rightmost nodes and $k/2$ leftmost nodes.

Figure 3.7 is an example of a regular ring lattice network. Let us measure the local clustering coefficient for a node V in a regular lattice network consisting of N nodes.

1. Spread the lattice network into a linear graph with respect to a node V, such that the k neighbours are split into $k/2$ neighbours on each side of node V as depicted in Figure 3.8. We are now interested in measuring the number of edges among the neighbours of V (be careful! here we ignore those edges which connect the neighbours of V with V itself; there are k such edges).
2. **Left side of V:** Node V_1 would be connected to $k/2-1$ nodes to its right, all of which are also V's neighbours. Similarly, node V_2 would be connected to $k/2-1$ nodes to its right, all of which are also V's neighbours. The total number of edges contributed by all the nodes ($k/2$ such nodes) on the left of node V in measuring V's clustering coefficient would therefore be

$$\left(\frac{k}{2}-1\right)+\left(\frac{k}{2}-1\right)+\ldots\frac{k}{2}\text{ times}=\frac{k}{2}\left(\frac{k}{2}-1\right)$$

Source: Social Network Analysis (Book) by Tanmoy Chakraborty

Ring Lattice Network Model

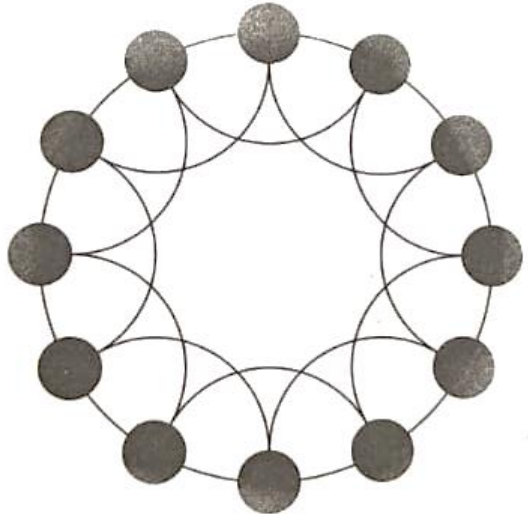


FIGURE 3.7. An example of regular ring lattice network with $k = 4$.

Source: Social Network Analysis (Book) by Tanmoy Chakraborty

Ring Lattice Network Model

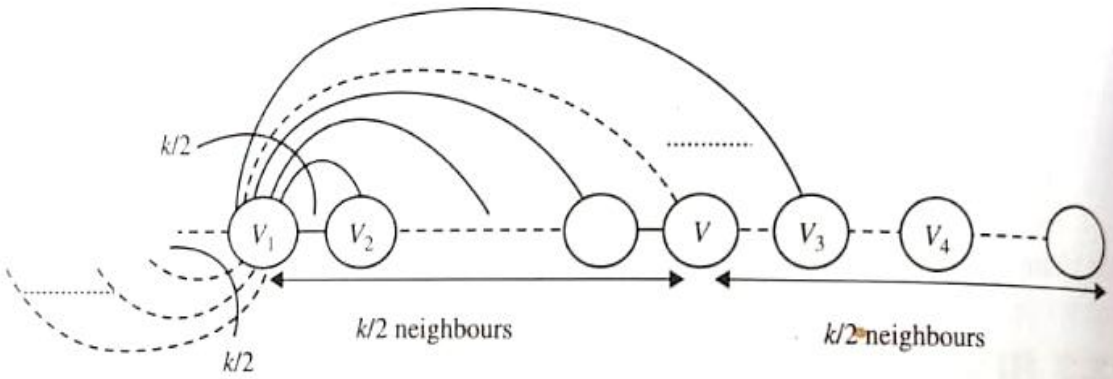


FIGURE 3.8. A lattice model spread out linearly with respect to a node V. There are $k/2$ neighbours on each side of the lattice network. In order to calculate the clustering coefficient of V, when we consider one of its neighbours V_1, we shall consider V_1's $k/2$ edges which are connected to V's neighbours. V_1's other links (dotted edges) are ignored in this case. In the same manner, we consider edges for V's other neighbours. This figure only shows V_1's edges contributing to the calculation of V's clustering coefficient.

3. **Right side of V**: Node V_3 would be connected to $k/2 - 1$ nodes to its right. However, node V_4 would be connected to $k/2 - 2$ nodes to its right (Why? Check how many nodes are remaining on the right-hand side.). Hence, the total number of edges contributed by all the nodes on the right of node V in measuring V's clustering coefficient would be

Source: Social Network Analysis (Book) by Tanmoy Chakraborty

Ring Lattice Network Model

3. **Right side of V**: Node V_3 would be connected to $k/2 - 1$ nodes to its right. However, node V_4 would be connected to $k/2 - 2$ nodes to its right (Why? Check how many nodes are remaining on the right-hand side.). Hence, the total number of edges contributed by all the nodes on the right of node V in measuring V's clustering coefficient would be

$$\left(\frac{k}{2} - 1\right) + \left(\frac{k}{2} - 2\right) + \cdots + 1 = \frac{\frac{k}{2}\left(\frac{k}{2} - 1\right)}{2}$$

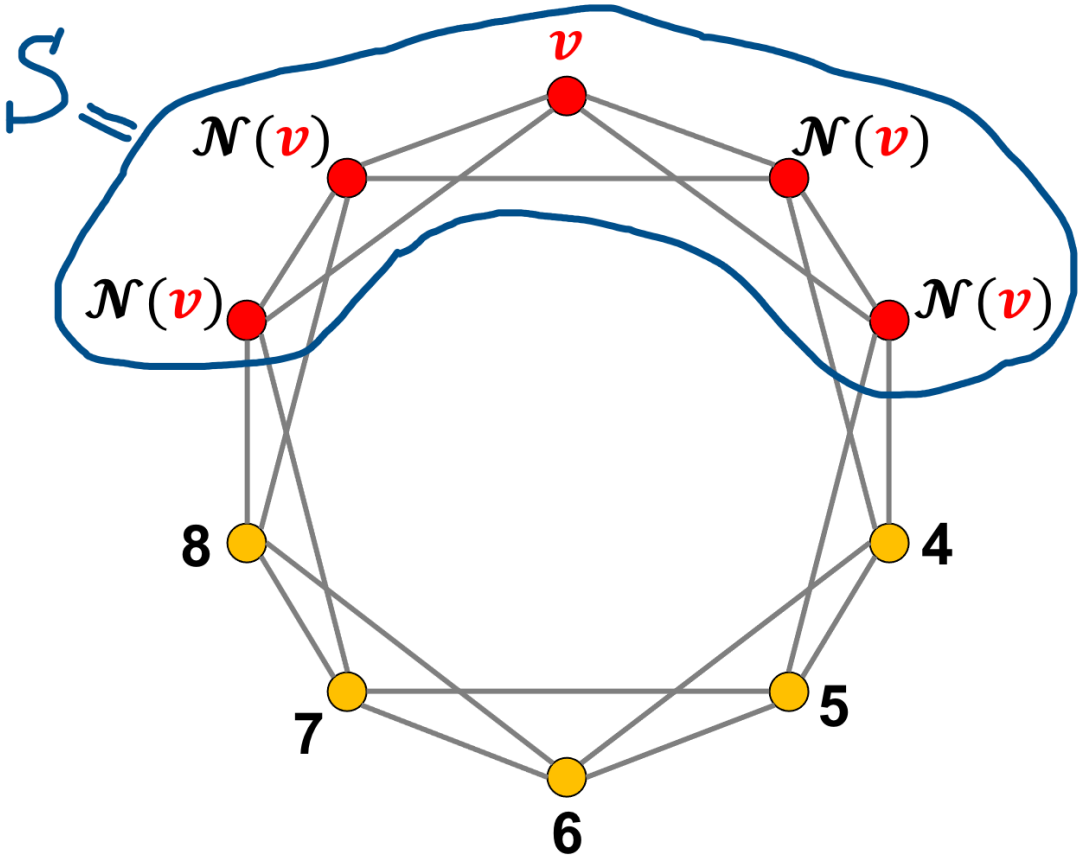
Adding the two expressions, we get the total number of edges among the neighbours of V as $(3/2)\,(k/2)(k/2 - 1)$. Total number of possible edges among k neighbours is $(k/2)(k - 1)$. Therefore, the local clustering coefficient C_V of node V as

$$C_V = \frac{\frac{3}{2}\frac{k}{2}\left(\frac{k}{2} - 1\right)}{\frac{k}{2}(k - 1)} = \frac{3(k - 2)}{4(k - 1)} \tag{3.9}$$

For a large k, we can approximate the above to $3/4$. Hence, for a large-lattice network, the local clustering coefficient is high. This mimics the behaviour of real-world networks, wherein despite large size of a network, we observe a high local clustering coefficient at many nodes.

Source: Social Network Analysis (Book) by Tanmoy Chakraborty

Ring Lattice Network Model: Local clustering coefficient



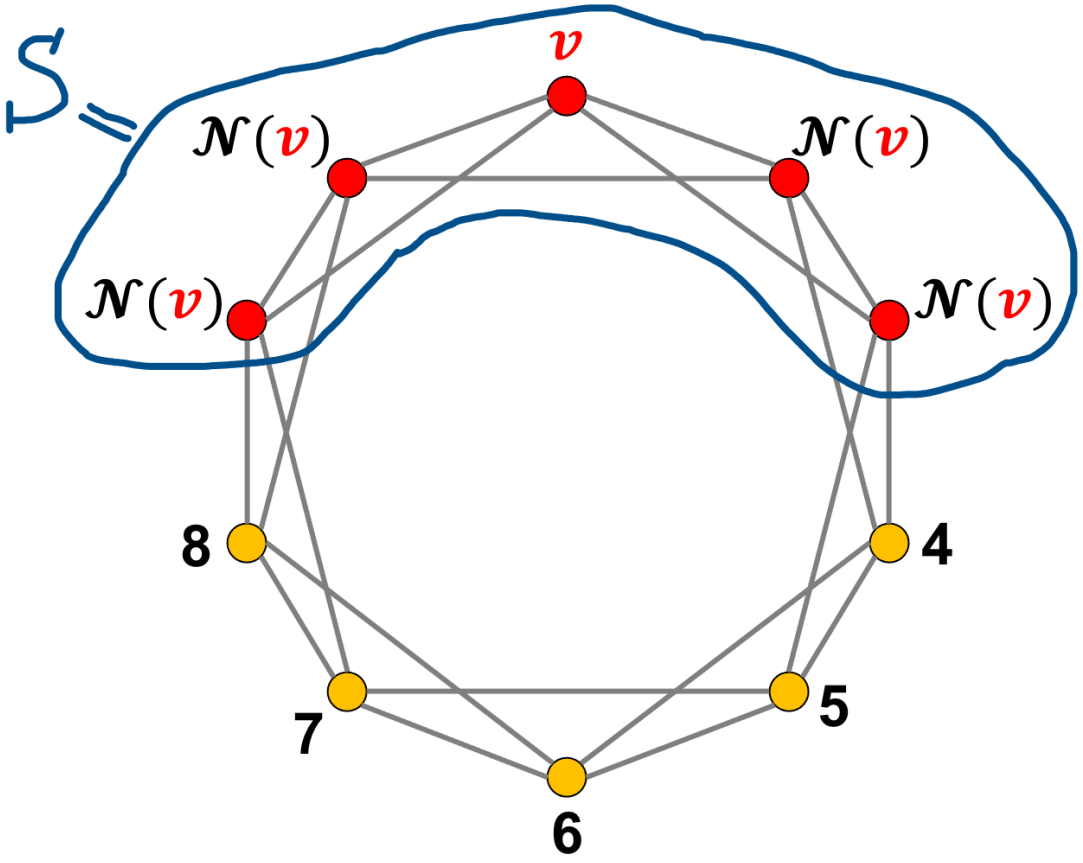
4-Regular Ring Lattice Network

- **Theorem:** Local clustering coefficient of any node v in a $k-Regular\ Ring\ Lattice\ Network$ is $\frac{3(k-2)}{4(k-1)}$

- When k is large, then the above expression boils down to $\frac{3}{4}$

- **Regular Ring Lattice networks has high local clustering coefficients**

Ring Lattice Network Model: small world property

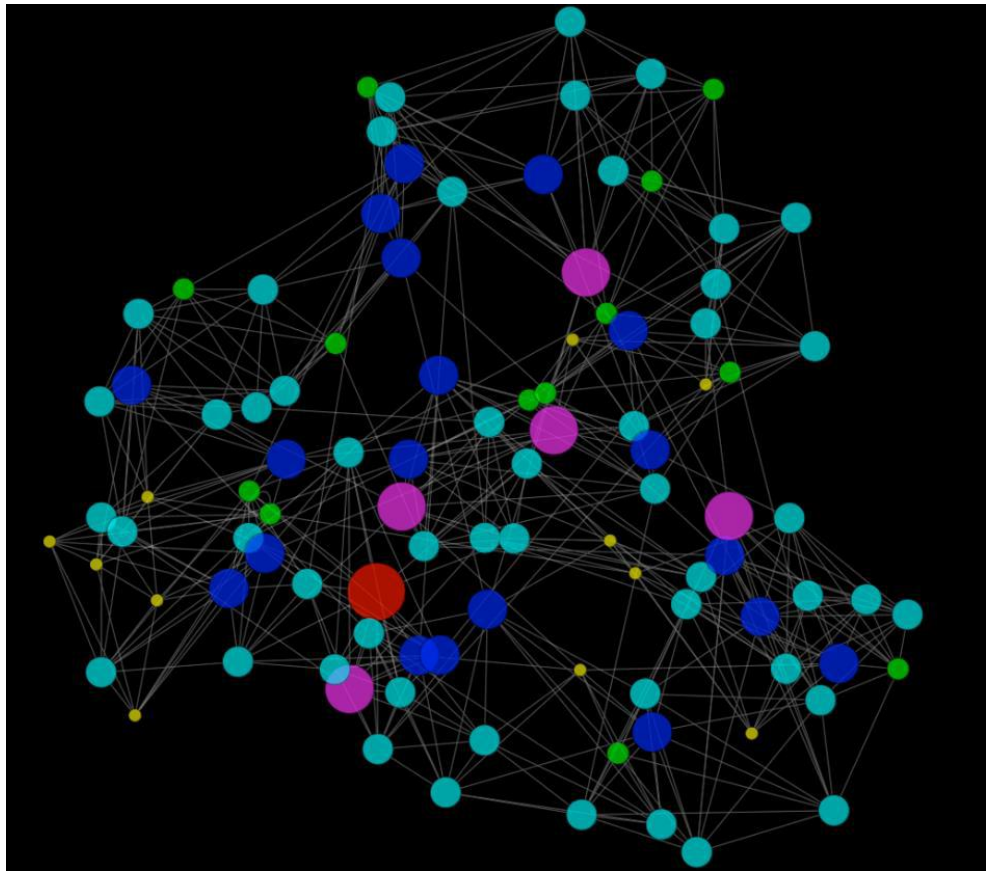


4-Regular Ring Lattice Network

- All the connection in ring lattice are to nearby nodes
- Two nodes that are in the opposite sides of the ring will be connected by long chains
- Regular Ring Lattice networks does not follow small-world property

Wattz-Storatz Model

Wattz-Storatz Model

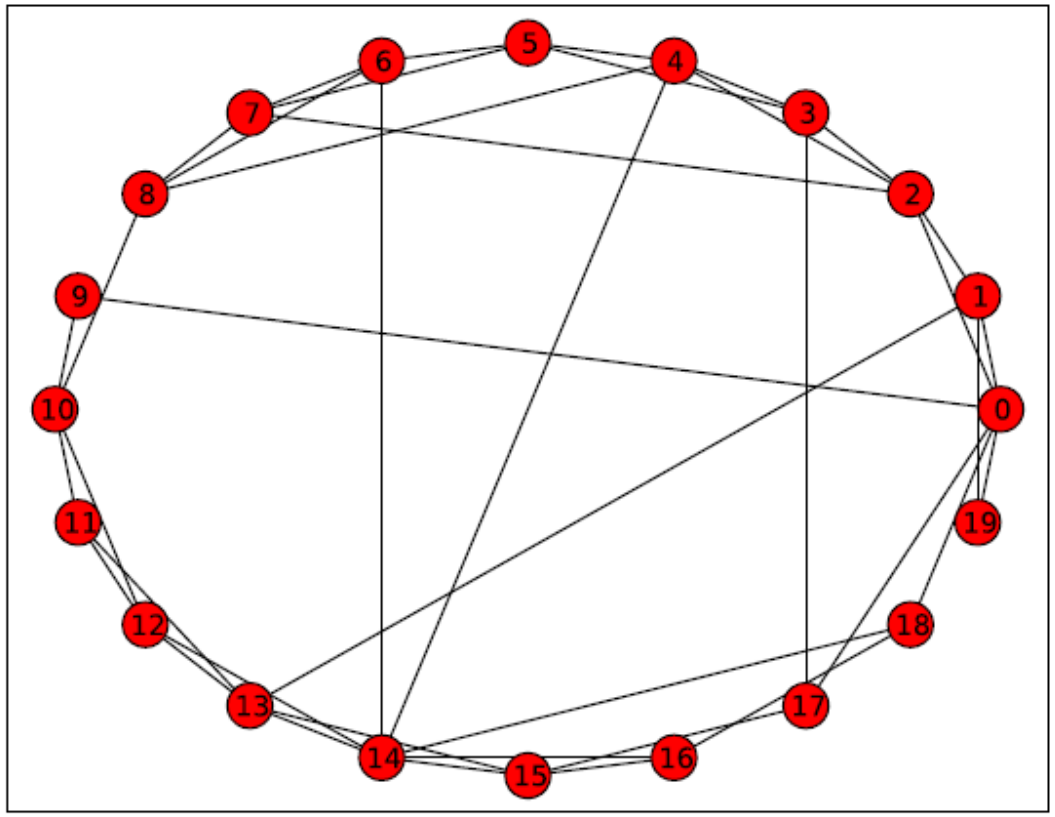


Watts–Strogatz network with 100 nodes formed by igraph and visualized by Cytoscape 2.5
https://en.wikipedia.org/wiki/Watts%E2%80%93Strogatz_model

- Erdős-RényiNetworks
  - possesses the small-world property
  - have small local clustering coefficients for all the nodes when the network size is large
- Regular Ring Lattice networks
  - possess high local clustering coefficients for all the node
  - does not follow small-world property
- Watts-Strogatz Model form networks that has a high local clustering coefficient and possesses the small-world property
- ❑Proposed by Watts and Strogatzin 1998

Wattz-Storatz Model: Network Formation

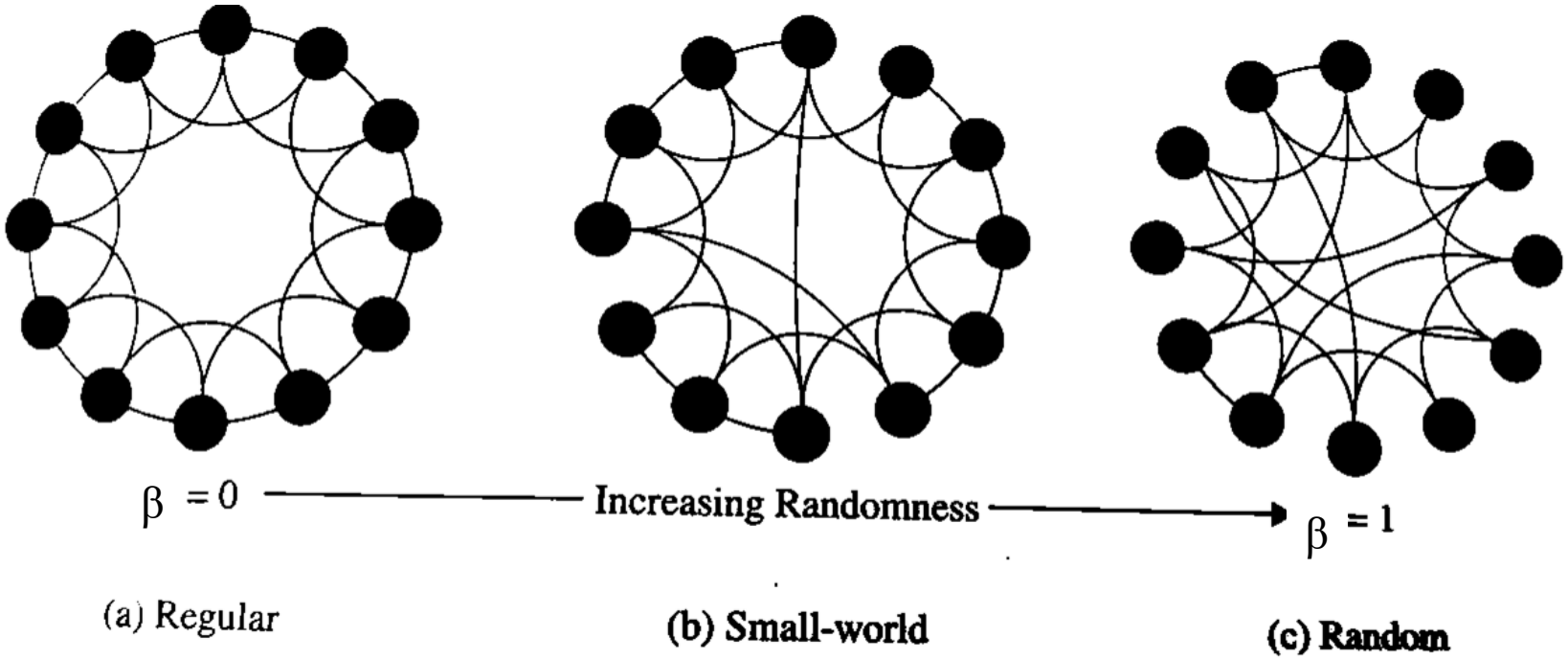
Watts-Strogatz model N=20, K=4, β=0.2



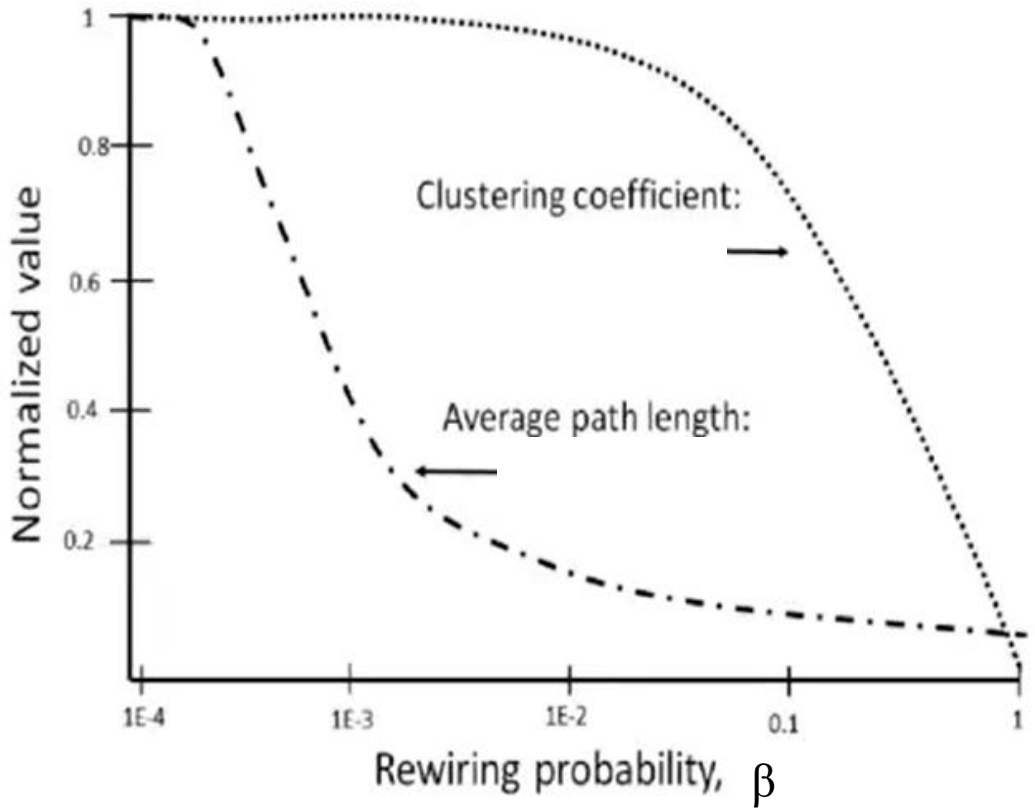
https://www.hindawi.com/journals/mpe/2014/693743/fig2/

1. Start with a k-regular lattice network of size N
2. List the nodes of the lattice as $1, 2, \cdots, N$
3. Choose i^{th} node from the list, $i = 1, 2, \cdots, N$
4. Select edges that link i^{th} node to some j^{th} node $(j > i)$
5. With a fixed rewiring probability β, rewire the other end of these edges
6. Avoid formation self-loops and link-duplication
7. Repeat steps 3 through 6 until all the nodes are scanned

Wattz-Storatz Model: Network Formation



Wattz-Storatz Model: Network Formation



- **Small rewiring probability**: When β is close to 0, the network has high clustering coefficient (~3/4) and high average path length as in lattice network.
- **Large rewiring probability**: When β is close to 1. the network has a low. clustering coefficient and a low average path length as in a random network.
- **Ideal rewiring probability**: the average path length, but the clustering coefficient remains high. Therefore, when $0.01 < \beta < 0.1$, we notice a behaviour that is similar to the real-world networks.

Wattz-Storatz Model: Observations

We have the following additional observations regarding the Watts-Strogatz network model:

1. No outliers: Similar to the random network, there are a very few outliers in the lattice network. This is unlike a real-world network, wherein there are significant outliers.
2. Small-world property: Lattice model does not follow the small-world property. In fact, the average path length scales linearly with time.
3. Power law: The degree distribution of the lattice model also does not follow power law. This is unlike real-world networks which are scale-free in nature.
4. Clustering coefficient: Similar to a real-world network, the lattice network has a high clustering coefficient.

Prefrential Attachement Model or Barabasi-Albert Model

Preferential Attachment Model or Barabasi-Albert Model: Background

- Hubs represent the most striking difference between a random and a scale-free network.
- On the World Wide Web, they are websites with an exceptional number of links, like google.com or facebook.com; in the metabolic network they are molecules like ATP or ADP, energy carriers involved in an exceptional number of chemical reactions.
- The very existence of these hubs and the related scale-free topology raises two fundamental questions:
  - Why do so different systems as the WWW or the cell converge to a similar scale-free architecture?
  - Why does the random network model of Erdős and Rényi fail to reproduce the hubs and the power laws observed in real networks?

Preferential Attachment Model or Barabasi-Albert Model: Background

The first question is particularly puzzling given the fundamental differences in the nature, origin, and scope of the systems that display the scale-free property:

- The *nodes* of the cellular network are metabolites or proteins, while the nodes of the WWW are documents, representing information without a physical manifestation.
- The *links* within the cell are chemical reactions and binding interactions, while the links of the WWW are URLs, or small segments of computer code.
- The *history* of these two systems could not be more different: The cellular network is shaped by 4 billion years of evolution, while the WWW is less than three decades old.
- The *purpose* of the metabolic network is to produce the chemical components the cell needs to stay alive, while the purpose of the WWW is information access and delivery.

- To understand why so *different* systems converge to a *similar* architecture we need to first understand the mechanism responsible for the emergence of the scale-free property.

Preferential Attachment Model or Barabasi-Albert Model

- **Growth and Preferential Attachment**
  - We start our journey by asking: Why are hubs and power laws absent in random networks?
  - The answer emerged in 1999, highlighting two hidden assumptions of the Erdős-Rényi model, that are violated in real networks.
    1) The random network model assumes that we have a *fixed* number of nodes, *N*. Yet, *in real networks the number of nodes continually grows thanks to the addition of new nodes.*
    2) The random network model assumes that we randomly choose the interaction partners of a node.

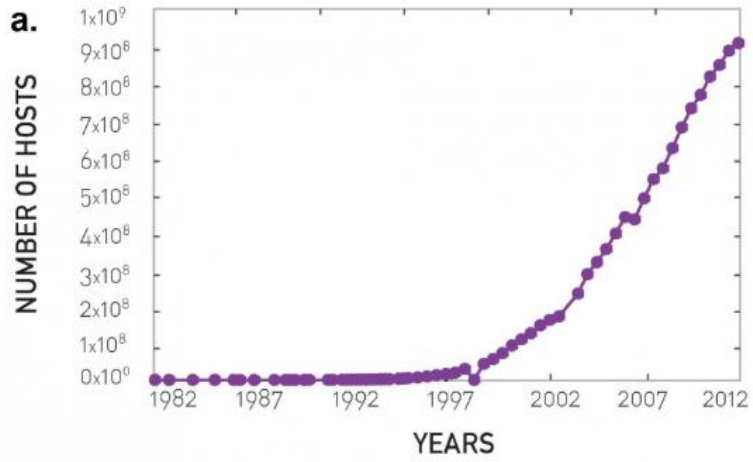
Preferential Attachment Model or Barabasi-Albert Model

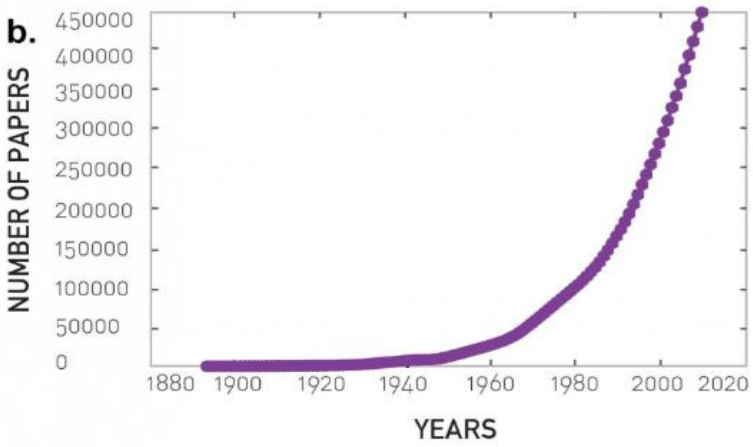
Networks Expand Through the Addition of New Nodes

The random network model assumes that we have a *fixed* number of nodes, *N*. Yet, *in real networks the number of nodes continually grows thanks to the addition of new nodes.*

Consider a few examples:

- In 1991 the WWW had a single node, the first webpage build by Tim Berners-Lee, the creator of the Web. Today the Web has over a trillion (10^{12}) documents, an extraordinary number that was reached through the continuous addition of new documents by millions of individuals and institutions (Image a).
- The collaboration and the citation network continually expands through the publication of new research papers (Image b).

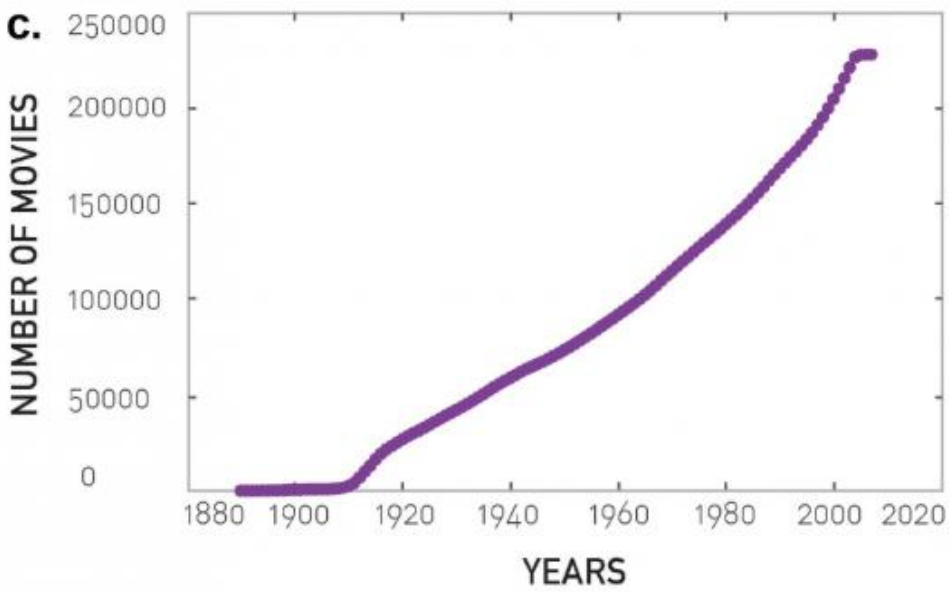




Preferential Attachment Model or Barabasi-Albert Model

- The actor network continues to expand through the release of new movies (Image c).
- The protein interaction network may appear to be static, as we inherit our genes (and hence our proteins) from our parents. Yet, it is not: The number of genes grew from a few to the over 20,000 genes present in a human cell over four billion years.

Consequently, if we wish to model these networks, we cannot resort to a static model. Our modeling approach must instead acknowledge that networks are the product of a steady growth process.



Preferential Attachment Model or Barabasi-Albert Model

Nodes Prefer to Link to the More Connected Nodes:

The random network model assumes that we randomly choose the interaction partners of a node. Yet, *most real networks new nodes prefer to link to the more connected nodes*, a process called *preferential attachment*

Consider a few examples:

- We are familiar with only a tiny fraction of the trillion or more documents available on the WWW. The nodes we know are not entirely random: We all heard about Google and Facebook, but we rarely encounter the billions of less-prominent nodes that populate the Web. As our knowledge is biased towards the more popular Web documents, we are more likely to link to a high-degree node than to a node with only few links.
- No scientist can attempt to read the more than a million scientific papers published each year. Yet, the more cited is a paper, the more likely that we hear about it and eventually read it. As we cite what we read, our citations are biased towards the more cited publications, representing the high-degree nodes of the citation network.
- The more movies an actor has played in, the more familiar is a casting director with her skills. Hence, the higher the degree of an actor in the actor network, the higher are the chances that she will be considered for a new role.

Preferential Attachment Model or Barabasi-Albert Model

In summary, the random network model differs from real networks in two important characteristics:

- **Growth**
  Real networks are the result of a growth process that continuously increases *N*. In contrast the random network model assumes that the number of nodes, *N*, is fixed.
- **Preferential Attachment**
  In real networks new nodes tend to link to the more connected nodes. In contrast nodes in random networks randomly choose their interaction partners.

There are many other differences between real and random networks. Yet, as we show next, these two, *growth* and *preferential attachment*, play a particularly important role in shaping a network's degree distribution.

The Barabási-Albert Model

The recognition that growth and preferential attachment coexist in real networks has inspired a minimal model called the *Barabási-Albert* model, which can generate scale-free networks. Also known as the BA model or the *scale-free model*, it is defined as follows:

We start with m_0 nodes, the links between which are chosen arbitrarily, as long as each node has at least one link. The network develops following two steps

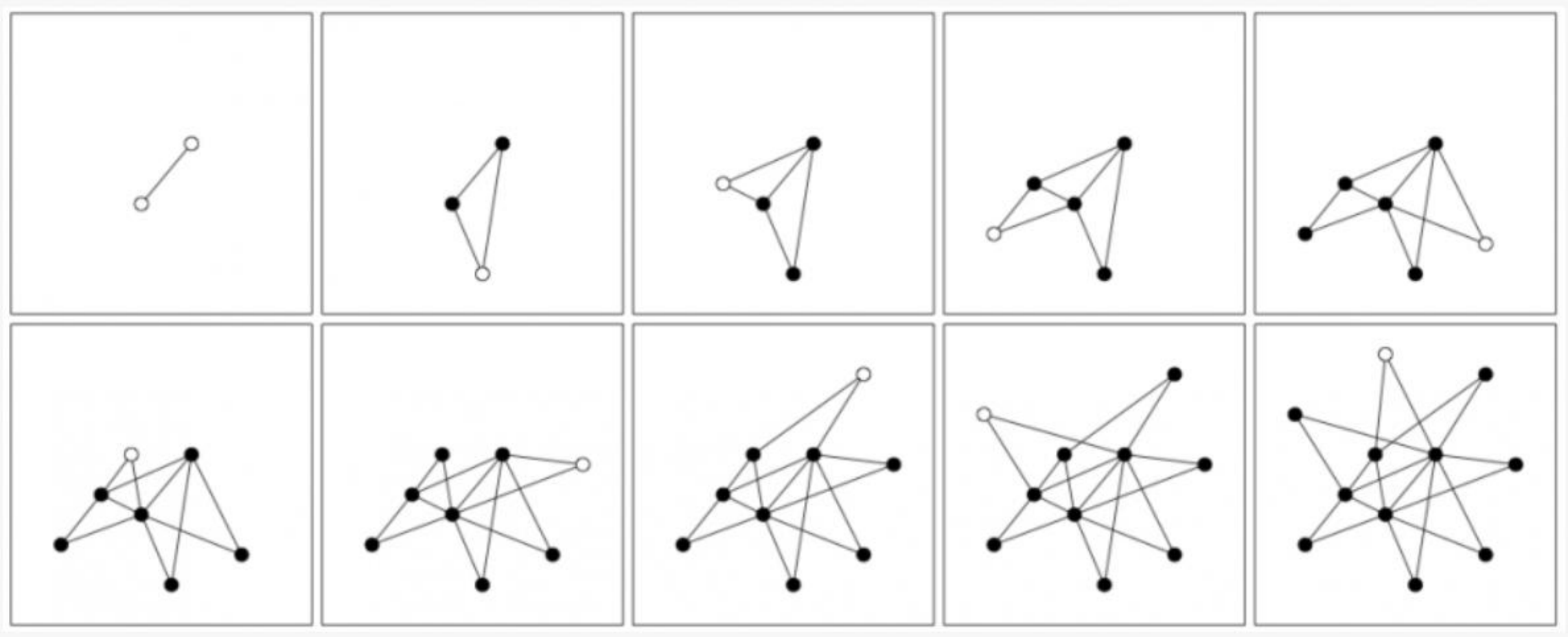
1. **Growth**
   At each timestep we add a new node with m ($\leq m_0$) links that connect the new node to m nodes already in the network.
2. **Preferential attachment**
   The probability $\Pi(k)$ that a link of the new node connects to node i depends on the degree k_i as

$$\Pi(k_i) = \frac{k_i}{\sum_j k_j} \qquad (5.1)$$

The Barabási-Albert Model

- Preferential attachment is a probabilistic mechanism: A new node is free to connect to *any* node in the network, whether it is a hub or has a single link. Equation (5.1) implies, however, that if a new node has a choice between a degree-two and a degree-four node, it is twice as likely that it connects to the degree-four node.

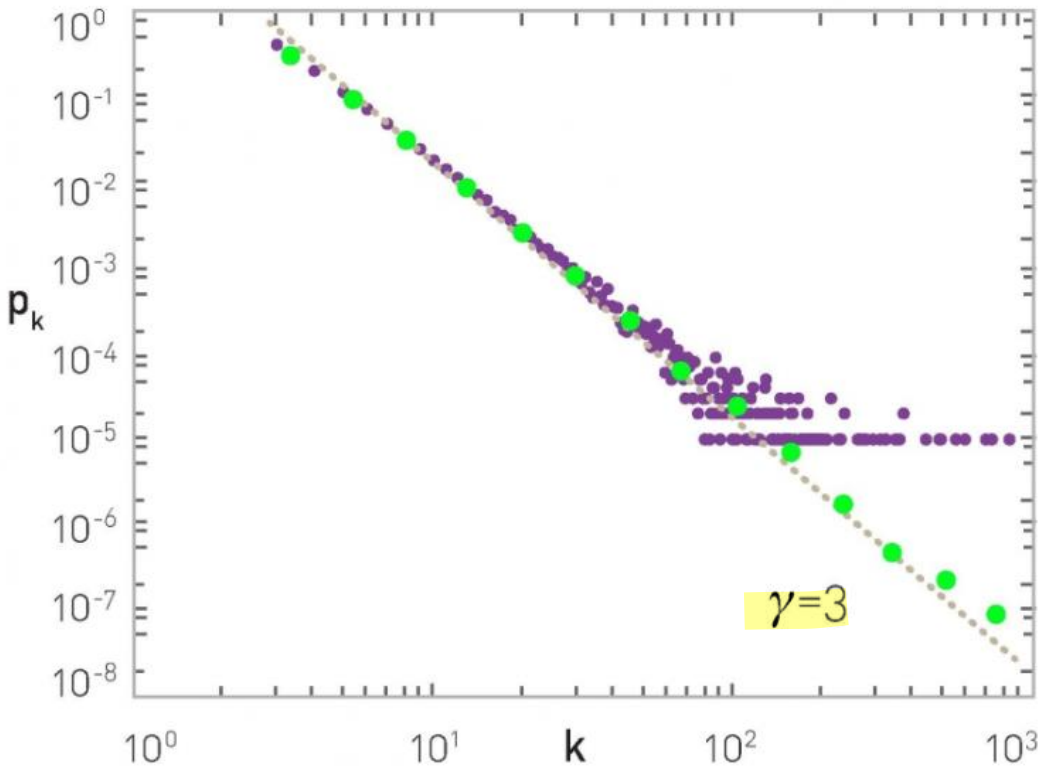
The Barabási-Albert Model



Evolution of the Barabási-Albert Model
The sequence of images shows nine subsequent steps of the Barabási-Albert model. Empty circles mark the newly added node to the network, which decides where to connect its two links (m=2) using preferential attachment (5.1).

The Barabási-Albert Model

- After t timesteps the Barabási-Albert model generates a network with $N = t + m_0$ nodes and $m_0 + mt$ links.
- As Image shows, the obtained network has a power-law degree distribution with degree exponent γ=3.
- The figure shows p_k for a single network of size N=100,000 and m=3. It shows both the linearly-binned (purple) and the log-binned version (green) of p_k. The straight line is added to guide the eye and has slope γ=3, corresponding to the network's predicted degree exponent.



The Barabási-Albert Model

- While most nodes in the network have only a few links, a few gradually turn into hubs. These hubs are the result of a ***rich-gets-richer*** *phenomenon*: Due to preferential attachment new nodes are more likely to connect to the more connected nodes than to the smaller nodes. Hence, the larger nodes will acquire links at the expense of the smaller nodes, eventually becoming hubs.
- In summary, the Barabási-Albert model indicates that *two simple mechanisms,* ***growth and preferential attachment****, are responsible for the emergence of scale-free networks*. The origin of the power law and the associated hubs is a *rich-gets-richer phenomenon* induced by the coexistence of these two ingredients. To understand the model's behavior and to quantify the emergence of the scale-free property, we need to become familiar with the model's mathematical properties.

The Barabási-Albert Model

- **Degree dynamics:**
  - To understand the emergence of the scale-free property, we need to focus on the time evolution of the Barabási-Albert model. We begin by exploring the time-dependent degree of a single node
  - In the model an existing node can increase its degree each time a *new* node enters the network. This new node will link to *m* of the *N(t)* nodes already present in the system. The probability that one of these links connects to node *i* is given by equation(5.1)
  - Let us approximate the degree k_i with a continuous real variable, representing its expectation value over many realizations of the growth process. The rate at which an existing node *i* acquires links as a result of new nodes connecting to it is

$$\frac{dk_i}{dt} = m\Pi(k_i) = m\frac{k_i}{\sum_{j=1}^{N-1} k_j} \qquad (5.3)$$

The Barabási-Albert Model

- **Degree dynamics:**
  - The coefficient *m* describes that each new node arrives with *m* links. Hence, node *i* has *m* chances to be chosen. The sum in the denominator of (5.3) goes over all nodes in the network except the newly added node, thus

$$\sum_{j=1}^{N-1} k_j = 2mt - m \qquad (5.4)$$

(recall that sum of all the degree = 2 × number of edges Because each edge contributes 2 degrees)

  - Therefore (5.4) becomes

$$\frac{dk_i}{dt} = \frac{k_i}{2t-1} \qquad (5.5)$$

The Barabási-Albert Model

- For large t the (-1) term can be neglected in the denominator, obtaining

$$\frac{dk_i}{k_i} = \frac{1}{2}\frac{dt}{t} \qquad (5.6)$$

- By integrating (5.6) and using the fact that $k_i(t_i)=m$, meaning that node *i* joins the network at time t_i with *m* links, we obtain

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{\beta} \qquad (5.7)$$

- We call β the *dynamical exponent* and has the value $\frac{1}{2}$

The Barabási-Albert Model

- **Observations from equation 5.7**
  - **Observation 1: Growth model follows power law**

$$K_i(t) \propto t^{\beta}$$

Which establishes that the growth model follows power law with respect to the time-stamp parameter.

  - **Observation 2: First mover advantage**

$$K_i(t) \propto \left(\frac{1}{t_i}\right)^{\beta}$$

Since t_i denotes the time of arrival of node v_i in the network, in preferential attachment model, the earlier the node joins the network (smaller t_i), the higher would be its degree. This phenomenon is known as the first-mover advantage.

- Slide Credits And Reference Material:
  - Social Network Analysis by Tanmoy Chakraborty
  - Network Science by Albert-László Barabási